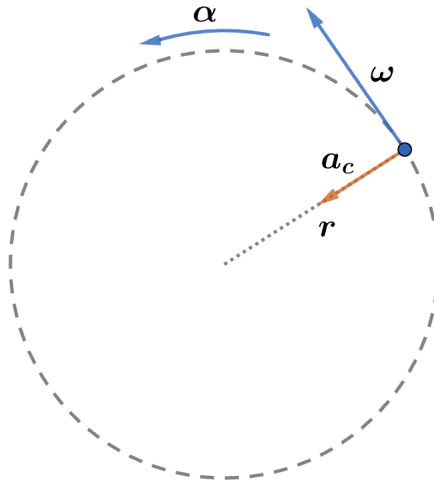


2014 F=ma Exam: Problem 10

Kevin S. Huang



Recall the centripetal acceleration is given by

$$a_c = \omega^2 r$$

Since the car starts at ω_0 and has angular acceleration α ,

$$\omega = \omega_0 + \alpha t$$

Substituting into the previous equation,

$$a_c = (\omega_0 + \alpha t)^2 r$$

We solve for the time t it takes the car to reach the critical centripetal acceleration $a_c = 243 \text{ m/s}^2$,

$$\sqrt{\frac{a_c}{r}} = \omega_0 + \alpha t$$

$$t = \frac{1}{\alpha} \left(\sqrt{\frac{a_c}{r}} - \omega_0 \right) = \frac{1}{4 \text{ s}^{-2}} \left(\sqrt{\frac{243 \text{ m/s}^2}{3 \text{ m}}} - 1 \text{ s}^{-1} \right) = 2 \text{ s}$$

so the answer is E.